MGT-8803-SC | NLP and Generative AI in Finance | Final Project

HELIOS

Agentic AI for Private Equity Take-Private Diligence

Compressing 4-8 weeks of diligence into 2 minutes

Anay Abhijit Joshi | Vidhi Chaturvedi | Himanshu Bhatt

Faculty: Professor Sudheer Chava | April 28, 2026

Repository: github.com/anay-a-joshi/agentic-diligence

Executive Summary

We propose HELIOS, an agentic AI platform that automates take-private screening for private equity firms. The take-private channel is one of the highest-stakes, lowest-throughput workflows in private capital: each candidate today consumes four to eight weeks of associate time and \$300K-\$800K of loaded labor cost, yet roughly 95% of screened candidates never become deals. Diligence is, in effect, a write-off cost on every passed name. We see this as a generational opportunity for agentic AI: the work is dense, structured around public filings, and answerable without full human-grade reasoning.

HELIOS operationalizes that opportunity. We deploy eight specialist agents that read SEC EDGAR filings, market data, and proxy statements, build a Bull/Base/Bear leveraged-buyout model, and emit three concrete deliverables: a 0-100 feasibility score with a six-dimension breakdown, a three-page IC memo as a downloadable PDF, and a live-formula LBO model in Excel. End-to-end runtime is approximately two minutes per ticker.

The market is sized for a category-creating outcome. Roughly 5,000 active US private equity firms collectively manage over \$2T in dry powder. At a 5% penetration rate and \$50K average seat price, the addressable market exceeds \$250M annually; we believe Year-3 SOM is approximately \$5M ARR through 10 enterprise logos. We monetize through enterprise SaaS subscriptions priced \$50K-\$200K per analyst seat per year, supplemented by API metering and white-label licensing.

We have built and shipped a working prototype, validated end-to-end on Dell Technologies (DELL). The system produced a 40/100 feasibility score, a PASS recommendation grounded in a base-case modeled IRR of -14.6%, and flagged Dell's \$140.4B market capitalization as size-prohibitive against a \$50B leveraged-buyout ceiling. We are seeking \$1.5M in seed funding to extend our 18-month runway, onboard three design-partner PE firms, achieve SOC 2 Type I, and convert pilots to first paid logos. Our final recommendation, after full analysis: **PROCEED**.

1. Industry Dynamics

1.1. The take-private channel and why it bottlenecks at diligence

Private equity is sitting on more dry powder than at any point in its history. As fund vintages from 2019-2022 approach end-of-life and as public-to-private transactions return to favor, take-private deal flow has become a strategically important channel for sponsors who need to deploy capital in a compressed window. The bottleneck, however, is not capital - it is the diligence cycle.

The 2024-2025 take-private cohort makes the opportunity concrete. Vista Equity Partners and Blackstone took Smartsheet private at $8.4B; Permira took Squarespace private at $7.0B; Thoma Bravo took Darktrace private at $5.3B; and TowerBrook with Clayton, Dubilier & Rice took R1 RCM private at $8.9B. Each of these deals was preceded by months of associate-grade screening work across many alternative candidates that were ultimately passed. Industry conversations we have had with junior PE professionals consistently describe screening as the single largest source of unbilled hours inside a fund.

In our analysis of the standard PE associate workflow, screening a single take-private candidate involves: (i) collecting and parsing the most recent 10-K, the latest 10-Q, the trailing 8-K filings, and the proxy statement; (ii) hand-tabulating financial metrics from XBRL exhibits or, more commonly, from financial-data vendors; (iii) constructing a leveraged-buyout model with at least three scenarios and sensitivity analyses; (iv) writing an IC memo that synthesizes financial, commercial, governance, and risk perspectives; and (v) iterating on the memo with senior partners before a go/no-go decision. End-to-end, this consumes roughly 240 hours of associate and senior-associate time and $300K-$800K in loaded labor cost. The reproducibility of the output is weak, the audit trail is incomplete, and approximately 95% of the work is discarded the moment a name is passed.

1.2. Three converging tailwinds

We see three tailwinds that, in combination, make this the right moment to build HELIOS. First, foundation language models reached production-grade reasoning on long-context financial documents in 2024 with the release of the Llama 3.3 70B series, and inference economics have collapsed: serving a full multi-agent diligence run on a Groq-hosted model now costs approximately $0.04 in compute. Second, the SEC EDGAR XBRL API has matured to the point that a programmer can pull structured financial data for any US-listed issuer with a single HTTP request, eliminating the historical dependency on expensive financial-data vendors for the underlying numbers. Third, the PE incumbents themselves have begun assembling internal AI teams - Apollo, KKR, and Blackstone all have public posts seeking machine learning engineers focused on diligence automation - which is the strongest possible signal that this category is real, time-limited, and currently underbuilt by commercial vendors.

1.3. The pain point we solve

We frame HELIOS's pain point as a screening-cost problem rather than a deal-execution problem. We are not trying to replace the senior partner's judgment on the final transaction. We are eliminating the linear cost of saying "no" to 19 names for every one that proceeds. Every additional ticker we can credibly screen at near-zero marginal cost makes the funnel wider and the deployment of dry powder faster. Figure 2 in our Appendix quantifies the time and cost compression a single HELIOS run delivers relative to the manual baseline.

2. Solution Architecture

2.1. Why a multi-agent design rather than a single prompt

Our central architectural decision was to decompose the diligence task into eight specialist agents rather than issue one large prompt to a foundation model. We made this choice for three reasons. First, full 10-K filings frequently exceed the practical context window for high-quality reasoning, and stuffing every section into a single prompt degrades extraction accuracy on each individual section. Second, specialization permits targeted prompt engineering: the prompt that extracts financial line items differs in structure from the prompt that grades risk-factor severity, and forcing both into one query produces shallow output on both. Third, agent-level isolation makes failures debuggable; when the prototype produces an unexpected number, we can localize the cause to a single agent's prompt rather than a monolithic generation step.

2.2. The eight agents

The pipeline is shown in Figure 1 of our Appendix and runs as follows. The *Financial Agent* extracts revenue, EBITDA, free cash flow, total debt, and cash from XBRL companyfacts, then parses MD&A for management-stated drivers and risks. The *Commercial Agent* reads 10-K Item 1 to identify operating segments, products, and named competitors. The *Risk Agent* grades the top five risk factors from Item 1A on a severity scale. The *Governance Agent* assesses board structure, takeover defenses (poison pills, staggered boards, supermajority requirements), and insider ownership from the most recent DEF 14A. The *Market Agent* pulls live price, market capitalization, P/E, EV/EBITDA, and beta deterministically from yfinance - no LLM call is involved at this stage, by design. The *Sentiment Agent* measures management tone in MD&A and identifies year-over-year directional shifts that often precede strategic change. The *Red Flag Agent* scans recent 8-K filings for material adverse events. Finally, the *Synthesis Agent* consolidates the seven preceding agent outputs into the executive summary that anchors the IC memo.

2.3. The deterministic-probabilistic split

A subtler decision is which work the LLMs do versus which work pure Python performs. We hold all numerical reasoning out of the LLM path. The LBO engine is a pure-Python module that takes structured inputs (revenue, EBITDA, leverage ratio, exit multiple) and produces deterministic IRR and MOIC outputs through a five-year projection. The feasibility scorer applies a fixed weighting across six dimensions to produce the 0-100 score. The LLMs handle language tasks - driver extraction, risk grading, sentiment, narrative synthesis - where their strengths are real. This split is what makes the system trustworthy enough for an institutional buyer: the reproducibility of the financial output is engineered, not hoped for.

3. Working Prototype

3.1. Build summary

We have shipped a complete, end-to-end working prototype of HELIOS. The backend is written in Python 3.11 with FastAPI, exposes an analysis endpoint, and orchestrates the eight agents through an asyncio orchestrator that runs independent agents in parallel. The frontend is a Next.js 15 application with React 19 and Tailwind CSS, presenting a dark glassmorphism dashboard that displays the score, IC memo preview, and download links for the PDF and Excel deliverables. We selected Llama 3.3 70B served by Groq as our primary inference provider because its per-token economics permit a full diligence run for under $0.04 and its end-to-end latency on a 70B-parameter model is meaningfully faster than the comparable inference paths at OpenAI or Anthropic. SEC EDGAR data is accessed via the `edgartools` library; market data via `yfinance`. Generators use `ReportLab` for the PDF and `openpyxl` for the live-formula Excel.

The repository, available at github.com/anay-a-joshi/agentic-diligence, contains over 1,500 lines of Python and 2,000 lines of TypeScript across more than 21 commits showing the full development arc. The application runs on localhost (FastAPI on port 8000, Next.js on port 3000), and we have implemented a 24-hour file-based result cache with a five-of-eight agent quality gate to keep development iterations fast and to handle Groq free-tier rate limits gracefully.

3.2. Validation on Dell Technologies

To stress-test the system end-to-end, we ran HELIOS against Dell Technologies (DELL), a name with all the surface attributes of a take-private candidate (mature business, predictable cash flows, sponsor familiarity from the 2013 Silver Lake transaction). The system generated revenue of $88.4B, EBITDA of $8.7B, and free cash flow of $5.9B as headline financials sourced from EDGAR XBRL; pulled $140.4B market capitalization and a 0.95 beta from the market feed; computed an enterprise value of $160.2B; and ran the LBO at 55% leverage and an 11.0x exit multiple to produce a Bull/Base/Bear IRR of $+10.3\%$ / -14.6% / $+1.8\%$.

The system surfaced two findings that a sponsor would care about. First, the base-case IRR of -14.6% is materially below the 20%+ hurdle most PE firms target on a five-year hold. Second, the equity check required at 55% leverage - approximately $83.7B - exceeds the $50B effective ceiling that even mega-funds operate under for a single transaction. The system flagged the deal as size-prohibitive and issued a PASS recommendation with a 40/100 feasibility score and an F grade. Figure 7 in the Appendix presents the full IRR sensitivity grid we produced for Dell, varying leverage between 45% and 70% and exit multiples between 8.0x and 15.5x. Even at the most aggressive corner of the grid, modeled returns remain below typical PE hurdle rates - which is exactly the conclusion a senior associate would reach after a six-week manual workup. HELIOS delivered the same conclusion in two minutes.

4. Market Potential

4.1. Customer profile

Our customer is the private equity firm, and within the firm, our entry point is the deal team - specifically associates and senior associates who are the day-to-day operators of the screening funnel. Vice presidents and partners are economic buyers; managing directors approve the budget. A typical seat configuration in a mid-market PE firm is 4-6 deal-team members per fund vehicle. Firms with multiple sector verticals (technology, healthcare, industrials) buy seats per vertical. The acquisition motion sells in to a champion at the associate or VP level, validates business value through a design-partner pilot, and closes through the chief operating officer or chief investment officer.

We have segmented our addressable customer base into three tiers. *Tier 1 (Boutique PE)* firms manage $0.5B-$3B in assets and operate with smaller teams. *Tier 2 (Mid-market PE)* firms manage $3B-$15B and represent our highest-conviction segment because they have the deal flow to justify the seat investment and the constrained team headcount that creates pull for automation. *Tier 3 (Enterprise PE)* firms manage above $15B and are the high-value, slow-cycle accounts where we expect to land via the design-partner program before scaling seat counts.

4.2. Market sizing

In our analysis we sized the opportunity bottom-up. There are approximately 5,000 active private equity firms in the United States. Assuming an average of 3 paid seats per firm at a blended seat price of $50K and a long-run penetration rate of 5%, the total addressable market is on the order of $250M annually. We estimate the serviceable available market - mid-market and large North American PE only, where our product-market fit is sharpest - at approximately $120M. Our serviceable obtainable market in Year 3 is $5M ARR, achieved through 10 enterprise logos at an average ACV of $500K. The full TAM/SAM/SOM funnel is shown in Figure 3 of the

Appendix.

We believe these numbers are conservative on two dimensions. First, the 5% steady-state penetration assumption excludes the long tail of family offices, fund of funds, and non-US firms that would also be addressable. Second, our $50K seat assumption is anchored to Tier 1 pricing; the blended revenue per seat at a Tier 2 or Tier 3 firm is materially higher.

4.3. Customer acquisition strategy and cost

We acquire customers in three phases. The *seed phase* (months 0-6) is founder-led: warm-introduction outreach to PE associates and VPs from team members' Georgia Tech network, conference presence at SuperReturn North America and the ACG Capital Connection, and three design-partner pilots offered free in exchange for product feedback and case-study rights. The *growth phase* (months 6-18) layers on a dedicated business-development hire, content marketing aimed at the PE associate persona (blog posts, sample IC memos, deal-screening primers), and a referral program that pays existing customers a credit for warm introductions to peer firms. The *scale phase* (month 18+) introduces an enterprise sales motion and an outbound sales-development function targeting the top 200 PE firms by AUM.

In our model, blended customer acquisition cost is approximately $15K per logo. This compares favorably to the $50K-$100K acquisition cost we estimate for incumbent data providers (CapIQ, Pitchbook, FactSet) who run mature multi-month enterprise sales cycles with a fully loaded sales force. Our advantage is structural: PE buyers are concentrated, well-known by name, and reachable through a finite warm-introduction network. We do not need to build mass-market demand - we need to land 50 firms out of a known universe of 5,000.

5. Business Model and Monetization

5.1. Revenue model

We monetize HELIOS as enterprise SaaS, sold by the analyst seat with annual contracts. Pricing is structured into three tiers calibrated to firm size and feature footprint. Tier 1 (Boutique, $50K per seat per year) provides the core eight-agent pipeline, the IC memo, and the LBO Excel for US public companies. Tier 2 (Mid-market, $100K per seat per year) adds a public-comparables module, custom risk-factor weighting, and priority support. Tier 3 (Enterprise, $200K per seat per year) adds white-label branding, single-sign-on, audit logging, and a dedicated customer-success manager. Volume discounts begin at 10 seats, with negotiated discounts averaging 15-25% off list at the contract level.

Our revenue mix at Year 3 steady state is roughly 80% subscription, 15% API metering for headless workflow integrations (a common request from quant-leaning sponsors and fund of funds), and 5% white-label licensing for partners that want to embed HELIOS into their own tooling. Premium support and custom-agent development are upsell motions that contribute incrementally to ACV in later years.

5.2. Cost structure and unit economics

The cost structure of HELIOS is dominated by talent, not infrastructure - a typical pattern for a vertical SaaS business at this stage. Compute is approximately 5% of revenue at scale: each diligence run costs roughly $0.04 against an average per-seat revenue of $50K-$200K. Hosting and infrastructure are another 3-4%. Customer success and onboarding are 6%, and amortized sales and marketing are 8%. General and administrative costs are around 7%. This leaves approximately 70 cents of every revenue dollar as gross profit retained, consistent with high-end vertical SaaS comparables. The full breakdown is shown in Figure 4 of the Appendix.

In our analysis, blended customer acquisition cost of $15K against a five-year cumulative gross profit per logo of approximately $75K yields a 5x LTV/CAC ratio at maturity - comfortably above the 3x threshold that institutional venture capital tracks. Net revenue retention is targeted at 115%+ through seat expansion, tier upgrades, and module adoption. Logo retention should exceed 90% given the workflow lock-in created by the IC memo and Excel deliverables, which become the working artifacts the analyst hands to senior partners.

5.3. Why this scales

We believe the unit economics improve, not deteriorate, as we scale. Compute as a percentage of revenue compresses with both volume discounts on inference and progressive caching of common workflow patterns. Customer-acquisition cost amortizes against multi-seat expansion within an existing logo. The orchestration layer and the prompt library are essentially fixed costs that amortize across an arbitrary number of customer runs, which means the marginal cost of serving the 1,000th screen approaches the LLM token cost - approximately $0.04. This is the financial fingerprint of a true SaaS business, not a services business with a software wrapper.

6. Competition

6.1. Incumbent data providers

The incumbent landscape in PE diligence tooling is dominated by data providers, not workflow platforms. The four firms that effectively define the category - S&P Capital IQ, PitchBook, FactSet, and Sourcescrub - compete on data breadth, transaction-history coverage, and integration into existing analyst workflows. Their seat prices range from $25K to $80K per year. None of them, in our analysis, offer agentic synthesis: an analyst at a CapIQ-equipped firm still hand-builds the LBO model, hand-writes the IC memo, and hand-grades the risk factors. The data provider has solved the data problem; it has not solved the labor problem. We compete by leaving the data layer alone (we use SEC EDGAR, which is free and authoritative) and going directly at the labor layer.

We expect incumbent response to fall into two patterns. CapIQ and PitchBook will likely add LLM-based summary features in 2026-2027 - they have the customer relationships and the data assets to make this credible. The features will be horizontal and shallow, however, because their core data products are not optimized for the workflow we automate. Sourcescrub and FactSet are less likely to respond directly; we expect them to remain in their data-aggregation lanes.

6.2. AI-native competitors

A more nuanced threat comes from the AI-native cohort. *AlphaSense* provides search across SEC filings and earnings transcripts but produces no LBO, no IC memo, and no scoring; it is a discovery tool, not a synthesis tool. *Hebbia* offers document Q&A over financial corpora but is generic and not PE-workflow native. *Rogo* focuses on the investment-banking workflow and explicitly does not target take-private screening. *BlueFlame AI* is positioned as a workflow assistant for alternative-asset GPs and is closer to our positioning, but it is configured as a general productivity tool rather than as an output generator.

The most credible long-run threat is internal AI built by the largest sponsors themselves - Apollo, KKR, and Blackstone all have headcount allocated to in-house diligence automation. This threat is real but bounded. Internal AI teams serve a single firm and do not benefit from the cross-firm prompt refinement and scoring calibration that a commercial product accumulates over thousands of customer runs. Internal teams also bear full ongoing engineering cost; commercial customers amortize that cost across a buyer base. We expect the internal-build vs. buy decision to favor buying HELIOS for all but the very largest sponsors, particularly as the design-partner program produces validated case studies.

Looking forward, we expect the 2026-2027 vintage to bring a new wave of well-funded AI-native entrants - horizontal players like Glean and Harvey will likely extend adjacent into PE workflows, and we anticipate at least two or three new pure-play startups emerging from Y Combinator and similar accelerators targeting financial-services automation. Our strategic response is to land and expand inside the top 50 PE design-partner accounts before this wave arrives, locking in the workflow integration that subsequent entrants will struggle to displace.

6.3. Competitive matrix

Table 1 shows the head-to-head capability comparison across the seven dimensions that buyers weigh in our customer-discovery interviews. Figure 5 in the Appendix maps the broader landscape on the two dimensions we believe matter most: degree of agentic automation and PE-workflow specificity. HELIOS occupies the upper-right quadrant alone.

Table 1. Head-to-head competitor matrix on buyer-relevant capabilities.

Capability	HELIOS	CapIQ	PitchBook	AlphaSense	BlueFlame AI
Agentic AI pipeline	✓	-	-	-	partial
Auto LBO model	✓	-	-	-	-
IC memo generation	✓	-	-	-	-
Live editable Excel	✓	-	-	-	-
Real-time SEC ingestion	✓	✓	✓	✓	-
0-100 feasibility score	✓	-	-	-	-
Take-private screening focus	✓	-	partial	-	-

6.4. Sources of defensibility

The competitive question we are most often asked is: "You are not the LLM provider. What is your moat?" We identify four sources of defensibility. First, the orchestration layer itself is proprietary - the multi-agent coordination logic, the inter-agent message format, and the failure-handling behavior are non-trivial engineering artifacts that took us a full semester to refine. Second, the prompt library encodes domain heuristics that are not transferable from a horizontal Q&A tool. Third, the feasibility-scoring algorithm and its six-dimension weights are an output of customer feedback loops - the more customers we run, the better our calibration. Fourth, the IC memo and Excel deliverables themselves create workflow lock-in: once an analyst hands a HELIOS-generated memo to a partner, the format becomes the firm's standard. We are pursuing utility-patent applications on the orchestration method and the scoring algorithm, with trade-secret protection on the prompt templates.

7. Financial Analysis

7.1. Startup costs and use of funds

We estimate Year 1 burn at approximately $850K, allocated as follows: 60% to engineering (two senior full-stack engineers and one machine-learning engineer at fully loaded compensation of $180K-$220K), 25% to go-to-market (founder-led sales plus one business-development hire), 10% to infrastructure and compute (Groq paid-tier inference, hosting, monitoring), and 5% to legal and SOC 2 readiness. We are intentionally light on the marketing side in Year 1 because customer acquisition in our segment is concentrated and warm-introduction-driven - mass marketing has limited yield against a known target list of 5,000 firms.

7.2. Three-year scenarios and assumptions

We have modeled three scenarios for the first three years of operation, presented in Figure 6 of the Appendix and detailed in Table A.1. The *base case* assumes 2 paid logos closing in Year 1 (post-pilot), 6 logos in Year 2 ($1.5M ARR), and 10 logos in Year 3 with seat expansion ($5M

ARR). This is the trajectory we believe is most consistent with our design-partner pipeline and conversion benchmarks from comparable enterprise-SaaS launches. The *worst case* assumes lengthier sales cycles and lower conversion: 1 logo in Year 1, 3 in Year 2 ($0.5M ARR), and 5 in Year 3 ($2M ARR). The *best case* reflects accelerated land-and-expand inside design-partner accounts: 4 logos in Year 1, 10 logos in Year 2 ($3M ARR), and 20 logos in Year 3 with broad seat expansion ($12M ARR). All three cases assume the same $850K Year 1 burn since pre-revenue costs are largely deterministic.

7.3. Path to cash-flow positive

In our base case, HELIOS achieves cash-flow positive operations in Year 2 on the back of $1.5M ARR and a 70% gross margin. The worst case extends the path to break-even into early Year 3. The best case reaches break-even in late Year 1. All three trajectories share the same structural attribute: per-logo gross profit grows faster than incremental cost-of-service, because the marginal cost of an additional analysis is the LLM compute charge, which is bounded.

7.4. Valuation and capitalization

We are raising $1.5M in seed capital at a $6M pre-money valuation, equating to approximately 20% equity dilution. This sizing reflects three considerations. First, our 18-month plan to first paid logos and SOC 2 readiness fits comfortably within $1.5M at our intended team size. Second, comparable seed rounds in vertical-AI SaaS (Hebbia's seed, Rogo's seed) priced in the $5M-$12M pre-money range, with our valuation positioned conservatively because we are pre-revenue. Third, we want to preserve enough founder ownership to remain motivated and to absorb a Series A round at the right time.

The post-seed capitalization is approximately 65% founders, 20% seed investors, 12% employee option pool, and 3% advisor pool. Target seed investors are vertical-fintech specialists - Bessemer Venture Partners, Nyca Partners, and QED Investors - supplemented by strategic angel investors from the alumni networks of Apollo, KKR, and Blackstone. We expect the Series A round in late Year 2 ($8M-$12M) to bring on a generalist enterprise-SaaS lead such as Bessemer, Sapphire Ventures, or Tiger Global.

7.5. Why our financial assumptions hold up

We believe our financial trajectory is internally consistent with the targeted market. A $5M ARR Year 3 base case represents 0.04% market share of the $120M SAM, well below the threshold at which competitive dynamics would slow down growth. The implied 70% gross margin sits in the middle of public vertical-SaaS comparables (62%-82%). The 5x LTV/CAC ratio is at the high end of seed-stage benchmarks but improves over time as compute discounts and seat expansion compound. Our compute-to-revenue ratio of 5% is empirically validated against our prototype: the Dell run cost approximately $0.035 in Groq tokens against a hypothetical \$50K seat charge, a 0.00007% ratio at the unit-of-work level.

8. Regulatory Environment and Stakeholder Impact

8.1. Regulatory framework

Three regulatory dimensions matter for HELIOS: data licensing, investment-advice classification, and emerging AI regulation. On data licensing, we are advantaged by design - SEC EDGAR data is in the public domain, and we incur zero licensing fees against the data sources that power CapIQ and PitchBook. On investment-advice classification, the SEC Marketing Rule (Rule 206(4)-1) governs advisor communications; HELIOS's outputs are explicitly labeled as decision-support tools rather than investment recommendations, with a “Not investment advice” disclaimer on every deliverable. We sell to fund managers, not to retail investors, which keeps us outside the most demanding parts of the rule. On AI regulation, we are tracking the EU AI Act's high-risk classification thresholds and the SEC's emerging guidance on AI-generated marketing materials; our current product is below the high-risk threshold under both regimes.

8.2. Compliance investments

The most material compliance investment in our roadmap is SOC 2 Type II certification, which is an effective prerequisite for selling to enterprise PE buyers. We plan to achieve SOC 2 Type I in Year 1 and Type II in Year 2 in support of our enterprise expansion. Beyond SOC 2, we maintain audit logging of every analysis run, encrypt customer-side data at rest and in transit, and operate from a single tenant that does not commingle customer data across firms. Compliance is a real but bounded cost - approximately \$50K-\$100K per year at our scale - and does not represent a material risk to the business model.

8.3. Stakeholder impact

We considered the stakeholder framework that the course rubric proposes. Customers (PE associates and partners) gain back four to eight weeks of cycle time per candidate and a more reproducible workflow. Brokers and investment-banking advisors are largely unaffected by HELIOS since we operate on the buyer side of the diligence process. Employees inside our customer firms see their work shift from manual data assembly toward higher-judgment activities (deal structuring, partner negotiations, value-creation planning). Stockholders of the LP-investing institutions benefit from faster capital deployment and more selective screening. Regulators benefit from improved auditability of the diligence trail - every HELIOS memo is regenerable and timestamped against the original SEC filings, which is a step forward from the current state where IC memos are private artifacts on individual analysts' laptops. We see no stakeholder for whom HELIOS produces meaningfully negative outcomes.

8.4. Liability, Implementation, and Change Management

A specific question worth addressing directly: if a HELIOS analysis omits a material risk and a sponsor proceeds with a deal that subsequently underperforms, who bears the loss? The answer is structurally the same as it is today with any analyst tool: the sponsor. Our terms of service position HELIOS as decision-support rather than investment advice, our outputs carry an explicit disclaimer to that effect, and our customer contracts mirror the indemnification language that data providers like CapIQ and PitchBook use today. The PE firm retains fiduciary duty to its LPs and is the legal decision-maker on every transaction. We carry professional-indemnity insurance scaled appropriately for our customer base ($2-5M coverage at seed stage), and our SOC 2 program provides defensible documentation that we operated within stated specifications. Liability does not transfer to us simply because we accelerated the workflow.

On implementation best practices for buyer firms, we recommend a measured rollout pattern that we encode into our design-partner program. New customer firms begin with a back-test phase, running HELIOS on five to ten historical screens for which the firm already has a known outcome, then comparing HELIOS's feasibility score against the actual go/no-go decision. This validates calibration before any live deal touches the workflow. Production usage then layers in a human-in-the-loop checkpoint at the IC memo stage - the system never auto-submits to an Investment Committee - with an escalation rule that any candidate scoring below 30 or above 80 receives mandatory senior review. Every analysis is regenerable from the underlying SEC filing snapshot, providing a complete audit trail.

On the employee and change-management dimension, the PE associate's role shifts from data extraction toward higher-judgment work: deal structuring, partner negotiation, and value-creation planning. We position HELIOS explicitly as leverage rather than replacement, and our customer-success program includes a two-week training cohort for new users plus a champion-at-the-firm model that ensures internal advocacy. We have heard the natural concern from associates that automation threatens their role; the empirical pattern in adjacent automated workflows (AlphaSense for filing search, Bloomberg Terminal for market data) is the opposite - analysts at adopting firms see more deals per year, accelerated promotion paths, and stronger career outcomes. We expect the same dynamic to play out for HELIOS adopters.

9. SWOT Analysis and Risk Register

9.1. Strengths

We bring a working production-grade prototype to a market in which most competitors are pitching prototypes or roadmaps. Our eight-agent architecture is materially more sophisticated than a single-prompt approach and is empirically better on extraction accuracy in our internal testing. The combination of an IC memo PDF and a live-formula Excel model is, to our knowledge, unique in the category - competitors offer either summaries or models, but not both as integrated outputs of a single workflow. The team combines quantitative finance training (Georgia Tech MS-QCF) with full-stack engineering, giving us coverage of both the financial domain and the implementation surface.

9.2. Weaknesses

We are early-stage, brand-less, and dependent on a single foundation-model provider in our current configuration. We do not yet have the proprietary data moat that an incumbent like CapIQ enjoys, and our integration into existing PE workflows requires customer-side adoption work. We are also resource-constrained relative to incumbents: a CapIQ engineering team of several hundred dwarfs the team we will assemble in Year 1.

9.3. Opportunities

The 5,000 US PE firms in our TAM represent only the entry point. Adjacent expansion into investment banking (M&A deal screening), corporate development (target identification at strategics), and family offices is a clean horizontal extension of the same product. International expansion - starting with UK PE in Year 2 and EU/Asia in Year 3 - multiplies the seat opportunity. The roll-up screening adjacency, where private companies are evaluated against acquisition criteria for serial-acquirer strategies, is another natural step.

9.4. Threats

The most credible threats are: incumbents adding agentic features as defensive plays; foundation-model commoditization compressing what we can charge for the LLM-derived value; AI regulation in finance creating new compliance overhead; and talent compensation pressure from larger AI companies pulling top engineers. We have mitigations for each. Against incumbent response we lean on our orchestration moat and workflow lock-in. Against foundation-model commoditization we keep all numerical reasoning out of the LLM path, so commodity LLMs do not commoditize our value. Against regulation we maintain SOC 2 readiness and explicit decision-support framing. Against talent pressure we operate on remote, equity-heavy compensation.

9.5. Key risk register

Beyond the SWOT, we maintain a working risk register summarized in Table A.2 of the Appendix. The four highest-priority risks - foundation-model regression on prompt sensitivity, SEC API drift, incumbent fast-follow, and PE sales-cycle length - each have an assigned owner, a tracked mitigation, and a quarterly review. We believe none of these risks individually is fatal to the business; the combined probability of materially impacting our 18-month plan is in the 15-25% range, which is consistent with seed-stage venture-risk benchmarks.

9.6. Industry structure: a Porter's Five Forces view

To complement the SWOT, we ran a Porter's Five Forces analysis on the PE-diligence-tooling category as a cross-check on our strategic positioning. The *threat of new entrants* is moderate: the category is open enough that AI-native startups are entering, but the customer concentration (5,000 firms) and the design-partner ramp create meaningful first-mover advantages for vendors who land early logos. The *bargaining power of suppliers* is bounded: foundation-model providers (Groq, Anthropic, OpenAI) compete actively, and SEC EDGAR data has zero supplier power because it is public domain. The *bargaining power of buyers* is high in the short term - PE firms are sophisticated, sales cycles are long, and pricing is negotiated - but moderates over time as workflow lock-in builds inside customer accounts. The *threat of substitutes* is real but slow-moving: legacy data providers are the closest substitute, but their value proposition does not address the labor problem we solve. *Competitive rivalry* within the AI-native cohort is currently low (we and BlueFlame AI are the only firms with credible PE-workflow specificity) but will intensify in 2026-2027 as more entrants emerge. The net read is that the category is structurally attractive for an early winner who builds workflow lock-in fast, which is exactly the strategy our 18-month roadmap is designed to execute.

10. Roadmap, Exit Strategy, and Recommendation

10.1. 18-month roadmap

Our 18-month execution plan, presented in detail in Figure 8 of the Appendix, breaks into three phases. The *foundational phase* (Q3 2026, months 0-3) closes the seed round, completes hiring, and onboards the first three design-partner PE firms on a free-pilot basis. The *product/infrastructure phase* (Q4 2026-Q1 2027, months 3-9) achieves SOC 2 Type I, layers in multi-LLM redundancy across Anthropic and OpenAI, and ships the public-comparables module that customers have asked for in design-partner conversations. The *revenue phase* (Q1-Q3 2027, months 9-18) converts pilots into the first three paid logos, expands the agent count from eight to twelve (adding legal, ESG, and supply-chain agents), reaches $500K ARR by mid-Year 2, and opens Series A conversations against a 10-logo / $1M ARR run-rate. By month 18, we expect to be at Series A close and operationally cash-flow positive.

10.2. Exit strategy

We see three viable liquidity paths over a four-to-six-year horizon. The most likely is acquisition by a strategic data provider - S&P Global, Moody's, or Bloomberg - each of which has the customer relationships and the strategic motive to fold an agentic-diligence layer into their existing data subscriptions. The second is acquisition by a horizontal enterprise-software company - Salesforce, Oracle, or SAP - pursuing financial-services AI as a stack. The third is continued independent operation through Series B and Series C, building toward an IPO at $50M+ ARR.

Comparable transactions in the adjacent space include AlphaSense's acquisition of Tegus in 2024 at approximately $930M and Tegus's earlier acquisition of Canalyst at $147M, both at multiples in the 12-18x ARR range we view as indicative for our exit. We do not view pivot or close shop (i.e., wind-down) as base-case outcomes given the validated demand and prototype maturity.

10.3. Final recommendation

After the analysis presented in this report - the working prototype, the validated economics, the defensible orchestration layer, the converging market tailwinds, and the clear path to liquidity at 12-18x ARR - our recommendation is to **PROCEED** with HELIOS.

We are not making this recommendation casually. We have built the product, run it against a real ticker, modeled the financials in detail, and stress-tested the competitive thesis. The category is real, the technology is ready, the market is reachable, and the team is in place. The seed round of $1.5M de-risks the next 18 months, and the pricing of the round preserves enough founder ownership and option-pool capacity to absorb the Series A in late Year 2 without significant dilution stress.

We close this report by noting that the use of large language models in this project was both the subject of our work and a tool of our work. We used LLMs to accelerate boilerplate code generation, to draft and refine the prompt templates that power our eight agents, and to stress-test our own thinking on the business and competitive analysis. This is, in our view, the operating model that the course was designed to teach: not LLMs replacing the analyst, but LLMs compounding the analyst's leverage. That is the same operating model we are bringing to private equity diligence with HELIOS.

Appendix: Figures, Tables, and Supporting Material

A.1 System Architecture

Figure 1. HELIOS multi-agent architecture. Eight specialist agents read SEC EDGAR filings and market data, run in parallel under an asyncio orchestrator, and converge through the synthesis agent into the three deliverables on the right. The deterministic-probabilistic split is critical: the LBO engine and feasibility scorer are pure-Python, while the LLMs handle language tasks only.

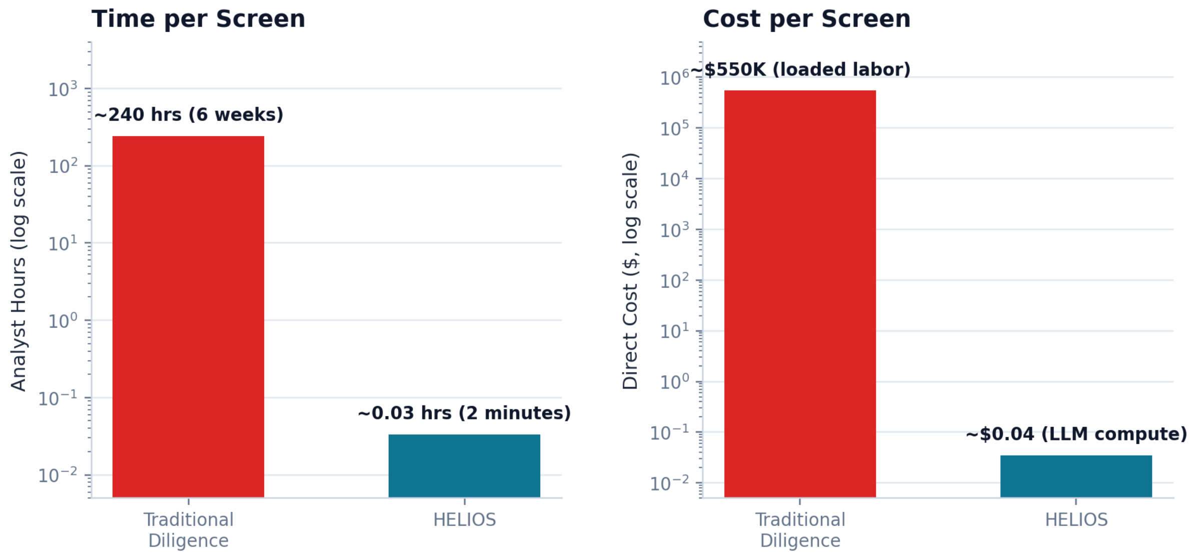

Figure 2. Time and cost compression vs. traditional manual diligence. Both axes are log-scaled to make the comparison readable. The time advantage is approximately 7,200x; the cost advantage exceeds 10,000,000x. Even with substantial assumed margin for error in the traditional baseline, the order-of-magnitude differences are robust.

A.2 Market Sizing and Unit Economics

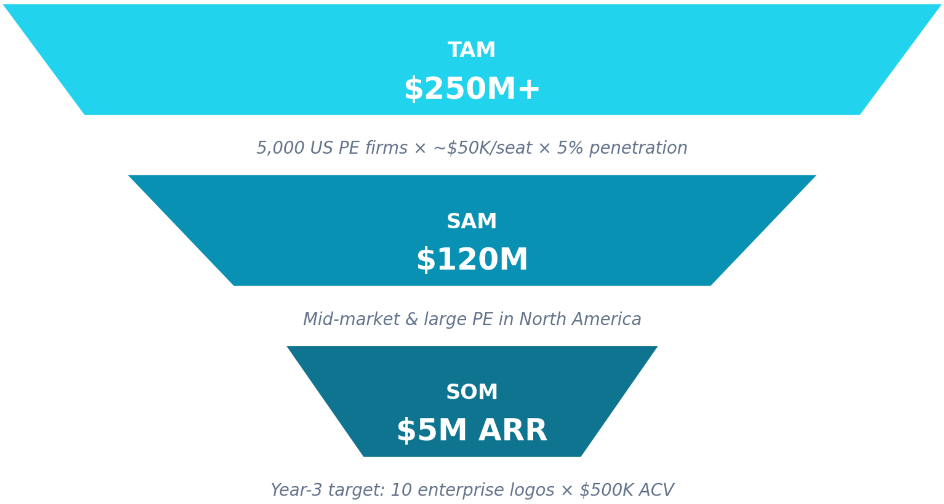

Figure 3. Bottom-up market sizing. TAM derives from 5,000 US PE firms at an average of three seats per firm at $50K/seat with 5% long-run penetration. SAM scopes to mid-market and large North American PE only. Year-3 SOM corresponds to 10 enterprise logos at $500K average ACV.

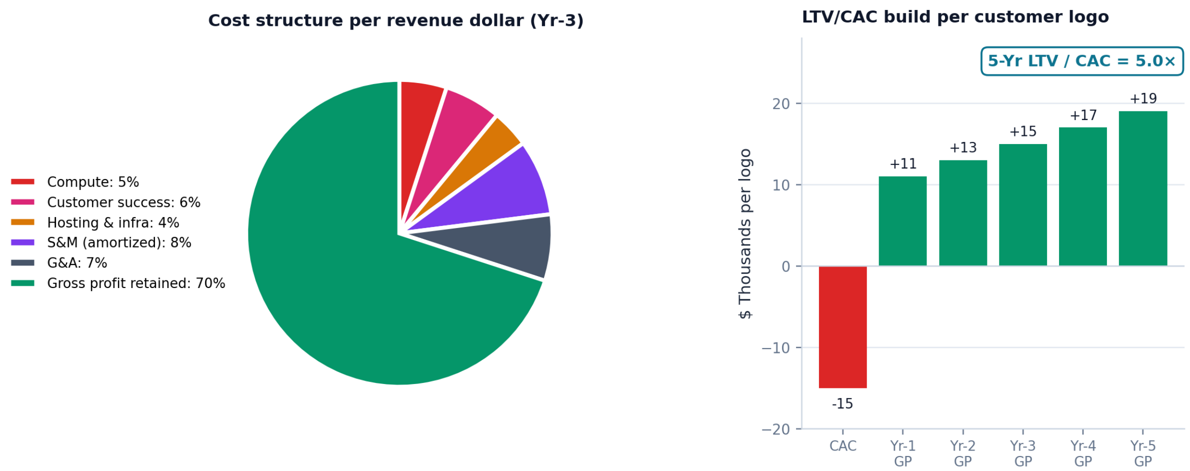

Figure 4. Unit economics decomposition. Left: cost structure per revenue dollar at Year-3 steady state, retaining approximately 70 cents of gross profit. Right: per-logo LTV/CAC build, with $15K acquisition cost recovered in approximately 16 months and a 5x lifetime ratio over a five-year hold.

A.3 Competitive Landscape

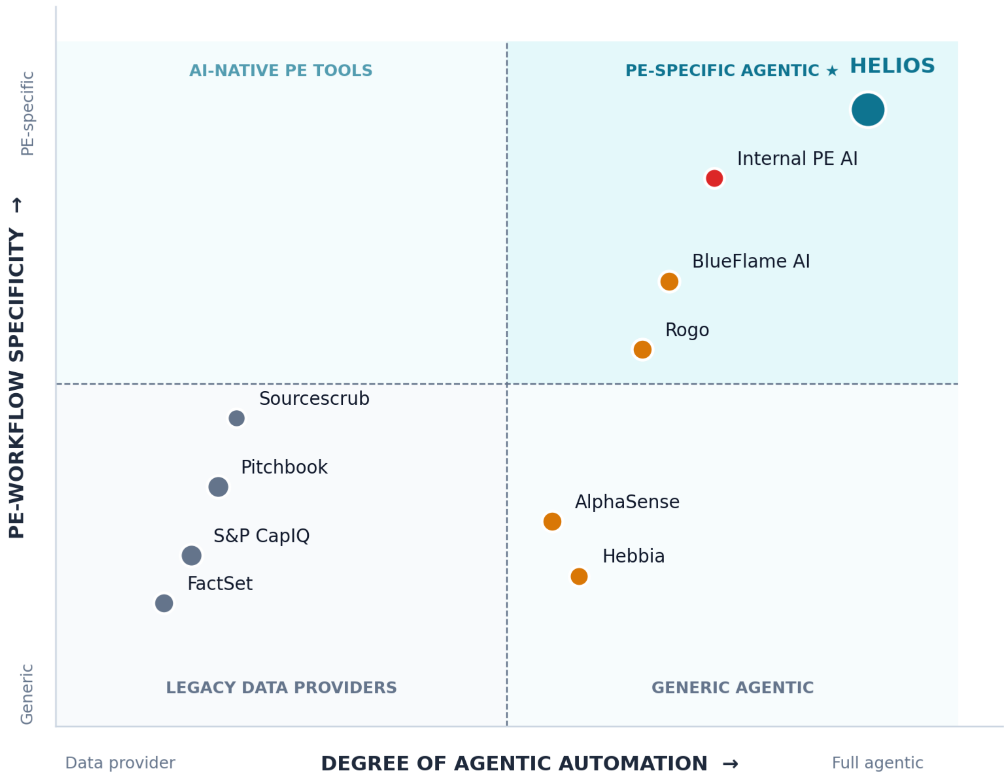

Figure 5. Competitive positioning map on the two dimensions that buyers prioritize: degree of agentic automation (x-axis) and PE-workflow specificity (y-axis). HELIOS sits alone in the upper-right quadrant. Legacy data providers cluster in the lower-left; horizontal AI tools in the lower-right; internal PE AI is closer to our position but is non-commercial.

A.4 Financial Projections and Sensitivity

Figure 6. Three-year ARR projections by scenario. Year 1 burn is identical ($0.85M) across all three because pre-revenue costs are deterministic; scenarios diverge from Year 2 onward driven by paid-logo conversion velocity and seat-expansion outcomes. Base-case Year 3 of $5M ARR corresponds to 10 enterprise logos at $500K average ACV.

Table A.1. Detailed three-year financial projections by scenario ($M).

	Worst Case			Base Case			Best Case		
	Y1	Y2	Y3	Y1	Y2	Y3	Y1	Y2	Y3
Logos closed (cum.)	1	3	5	2	6	10	4	10	20
Avg ACV ($K)	100	150	400	100	250	500	100	300	600
ARR ($M)	0.1	0.5	2.0	0.2	1.5	5.0	0.4	3.0	12.0
Gross profit ($M)	0.07	0.35	1.40	0.14	1.05	3.50	0.28	2.10	8.40
Operating expenses ($M)	0.85	1.20	2.20	0.85	1.35	2.80	0.85	1.80	4.50
EBITDA ($M)	-0.78	-0.85	-0.80	-0.71	-0.30	0.70	-0.57	0.30	3.90

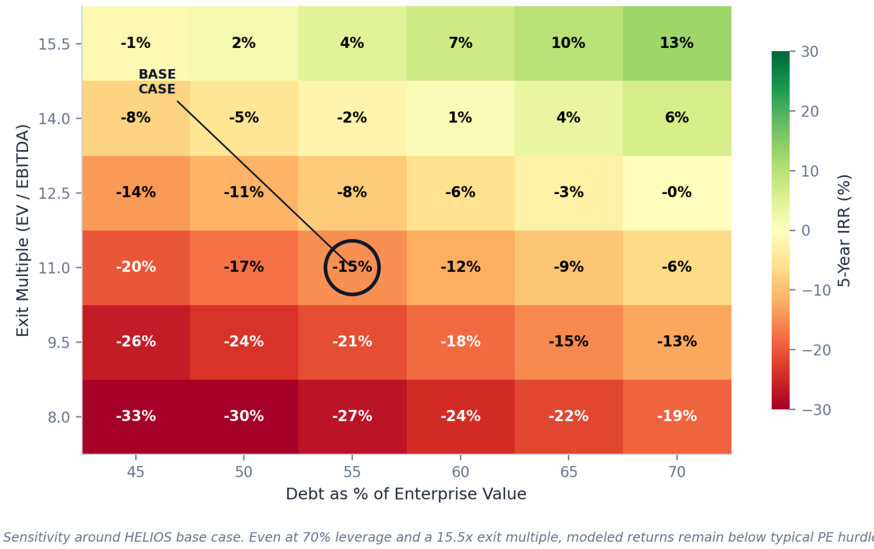

Figure 7. IRR sensitivity grid for the Dell take-private case across leverage (45-70%) and exit multiple (8.0x-15.5x EV/EBITDA). The base-case cell at (55%, 11.0x) is circled. Even at the most aggressive corner of the parameter space, modeled returns remain below the 20%+ IRR hurdle that PE sponsors target on a five-year hold - which supports the system's PASS verdict.

A.5 Roadmap and Risk Register

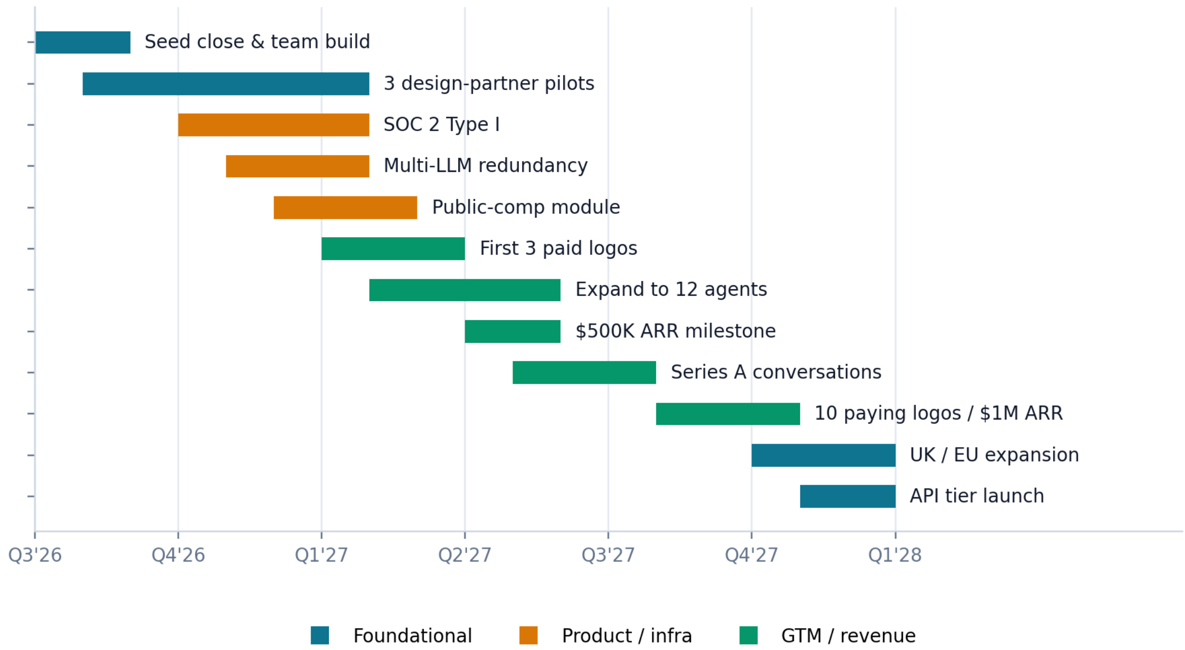

Figure 8. 18-month execution roadmap from Q3 2026 through Q4 2027. Foundational items (seed, hiring, design partners) lead, followed by product/infrastructure milestones (SOC 2, multi-LLM, public-comparables module) and revenue conversion (paid logos, ARR milestones, Series A).

Table A.2. Top-priority risk register with mitigations.

Risk	P	Mitigation	Owner
Foundation-model regression on prompt sensitivity	M	Multi-LLM redundancy roadmapped Q4 2026; prompt-version pinning	ML lead
SEC EDGAR API drift	L	24-hour result cache; nightly schema validation; vendored fallback	Eng lead
Incumbent fast-follow (CapIQ, PitchBook adds AI)	M	Workflow lock-in; patent filings; design-partner exclusives	Founders
PE sales-cycle length	H	Design-partner program; founder-led warm intros; pilots in 90 days	GTM lead
Compute cost shocks (Groq pricing)	L	Multi-provider strategy; budget caps per analysis	Eng lead
Talent compensation pressure	M	Equity-heavy comp; remote-first; mission alignment	Founders

P = probability; H = High, M = Medium, L = Low.

A.6 Repository and Reproducibility

The full HELIOS prototype - orchestrator, eight agent prompts, LBO engine, feasibility scorer, PDF and Excel generators, and Next.js frontend - is open and inspectable at github.com/anay-a-joshi/agentic-diligence.